

3.11 UPS

Uninterruptible Power Supplies are a class of power backup devices that meet the strict power requirements of PCs. PCs can tolerate a power outage without loss of data if power is restored within 10 milliseconds. If power is not restored within this time, the PC malfunctions causing a loss of data. A power backup system that meets this criteria can be called a UPS.



A UPS with an internal battery

3.11.1 AH

Short for Ampere Hour, this is an indication of the capacity of the battery of the UPS, and refers to the current that the battery can supply (in Amperes) in an hour. The higher the AH value, the longer the battery can power the system. The AH value should be governed by the expected backup period, and not the load offered by the system. There is little relation between the VA value of the UPS and the AH value of its battery. So a 600 VA UPS need not support a system longer than a 500 VA UPS if both of them use a 7.2AH battery for backup.

3.11.2 Battery

This refers to the secondary battery, usually sealed Lead Acid battery, used to store the charge to be used in the absence of utility power. Most 500 and 600 VA UPSs have a 7.2 AH battery inside. UPSes with external batteries also exist that allow additional batteries to be added to increase backup time. Larger UPSes use Lead Acid batteries as used by heavy vehicles for backup.

3.11.3 Line Interactive UPS

This type of UPS constantly checks the quality of the mains power. In case of any anomaly, like a reduction in voltage the UPS starts utilising the battery power and boosts the voltage going to the system. This system offers better protection than Offline UPS.